Active Learning #3
Ch 8-10

Class Reminders:

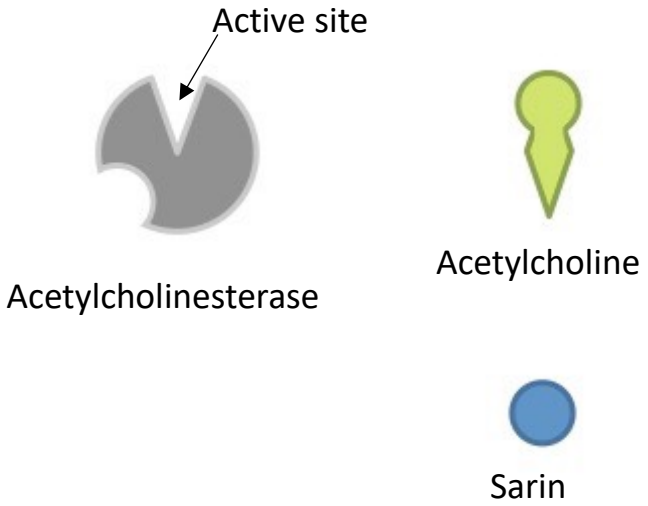
- Exam next class—please be on time! You only need to bring your UCF ID and a pencil.
- Ch 8-10 Post quizzes due Wednesday by 11:59pm.
- No class this Friday! (March 6)
- Final Exam Conflict Survey Due Friday (March 6)!
  - If you have a conflict with our scheduled final exam, this is your LAST opportunity to reschedule. You can find the link in our class Announcements.

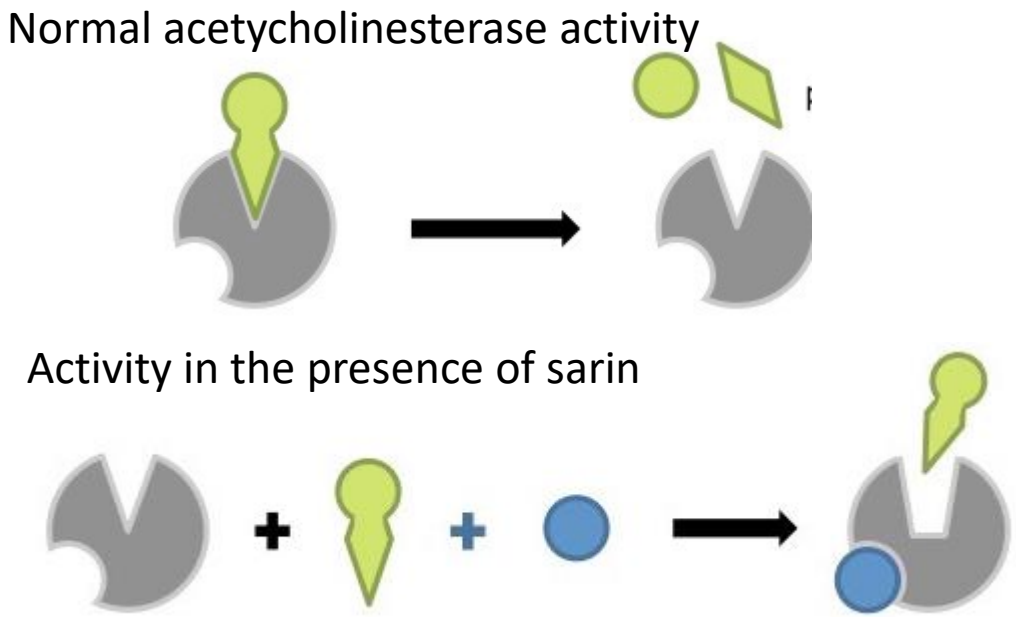
Reaction Coupling

- For A → B, ΔG = -200
  - Is the reaction A →B endergonic or exergonic?
  - Does it require energy or give off energy?
  - Which of the following reactions could be coupled with A → B?
    - C → D, ΔG = -350
    - E → F, ΔG = 375
    - G → H, ΔG = 250
    - I → J ΔG = 175
- For X → Y, ΔG = 75
  - Is the reaction X → Y endergonic or exergonic?
  - Does it require energy or give off energy?
  - Which of the following reactions could be coupled with X → Y?
    - C → D, ΔG = -125
    - E → F, ΔG = 225
    - G → H, ΔG = -50
    - I → J ΔG = 100

Enzyme Inhibition: Competitive or noncompetitive?

- Sarin Nerve Agent
  - Inhibitor for Acetylcholinesterase. Acetycholine is a neurotransmitter produced to induce muscle stimulation. When it is degraded, the muscle relaxes.

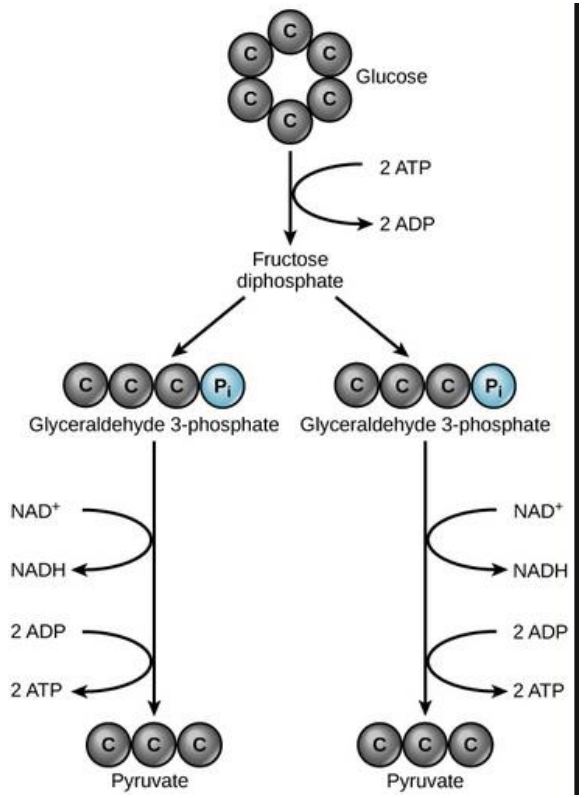




- What kind of inhibitor is Sarin?
- Based on shape, what kind of inhibitor would you expect Mullerian to be?



Cellular Respiration Overview: Glycolysis and Pyruvate Oxidation

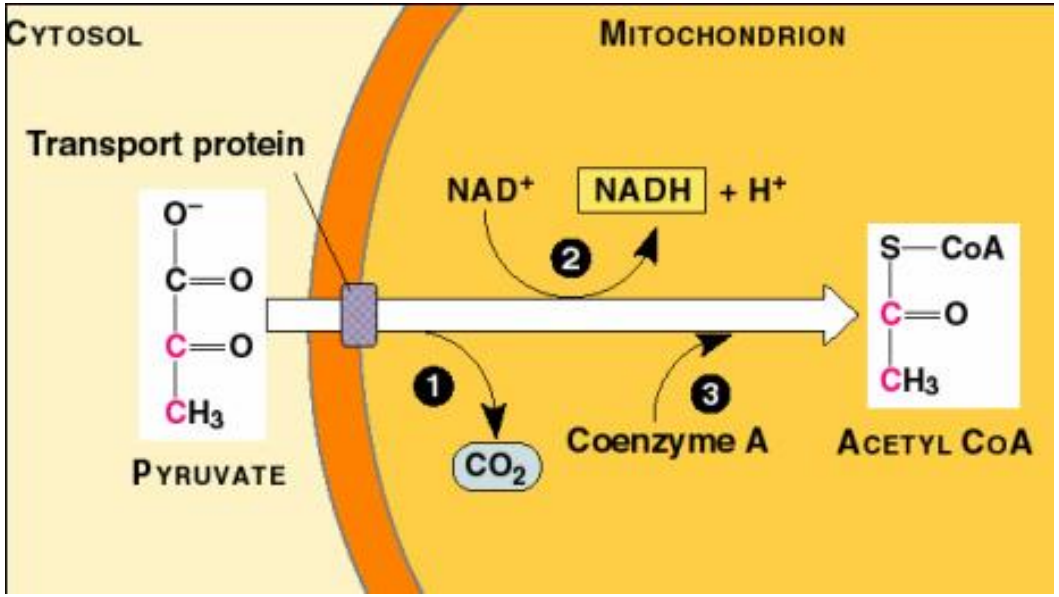


What are the products and what happens to them?

Product #1: _____ What happens to it? _____

Product #2: _____ What happens to it? _____

Product #3: _____ What happens to it? _____



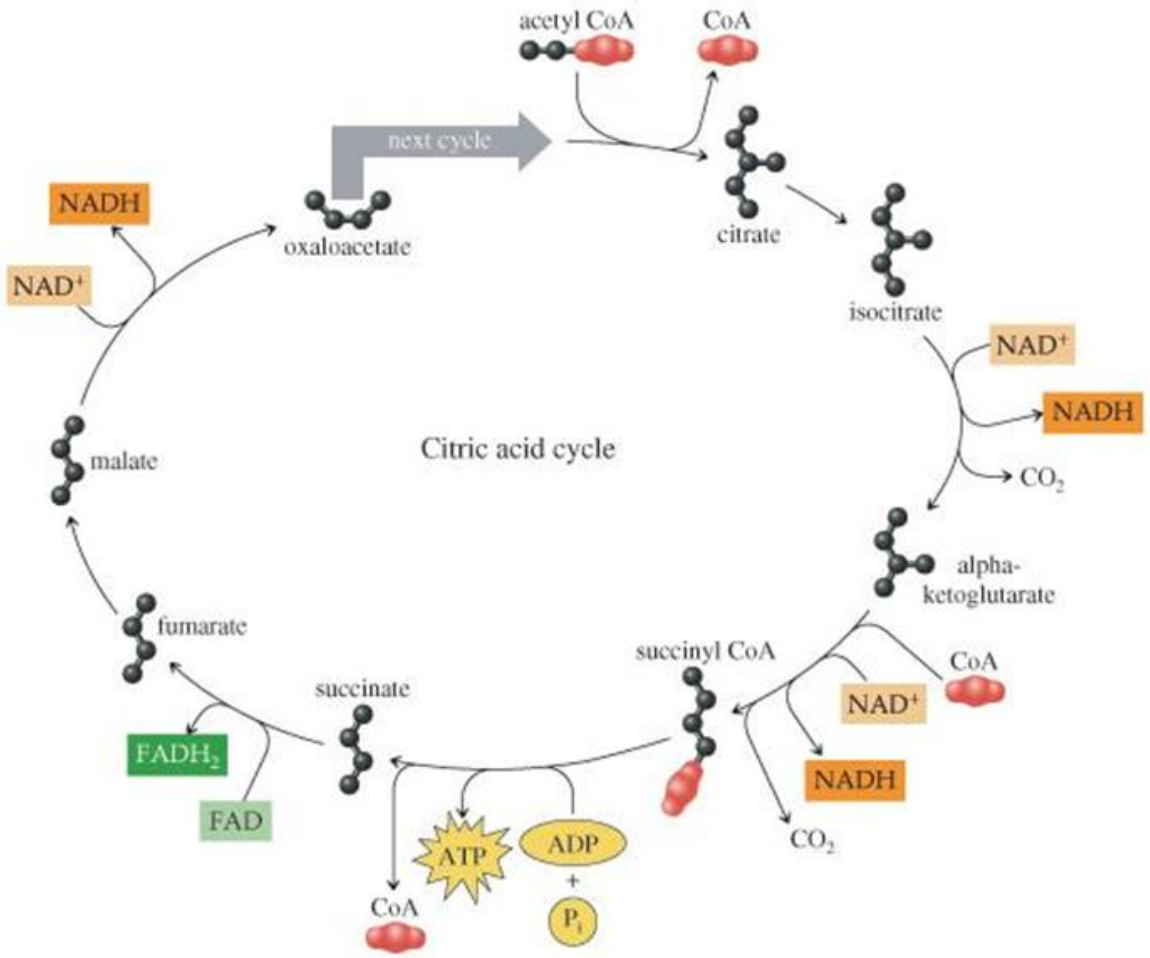
What are the products and what happens to them? (These answers are PER pyruvate. There are 2 pyruvates.

Product #1: _____ What happens to it? _____

Product #2: _____ What happens to it? _____

Product #3: _____ What happens to it? _____

Cellular Respiration Overview: The Citric Acid Cycle



This shows only one acetyl-CoA going through the CAC. Remember, there are two acetyl-CoA molecules and they each go through the CAC. So this happens twice.

What are the products and what happens to them? These answers are PER Acetyl-CoA. There are 2 Acetyl-CoA molecules.

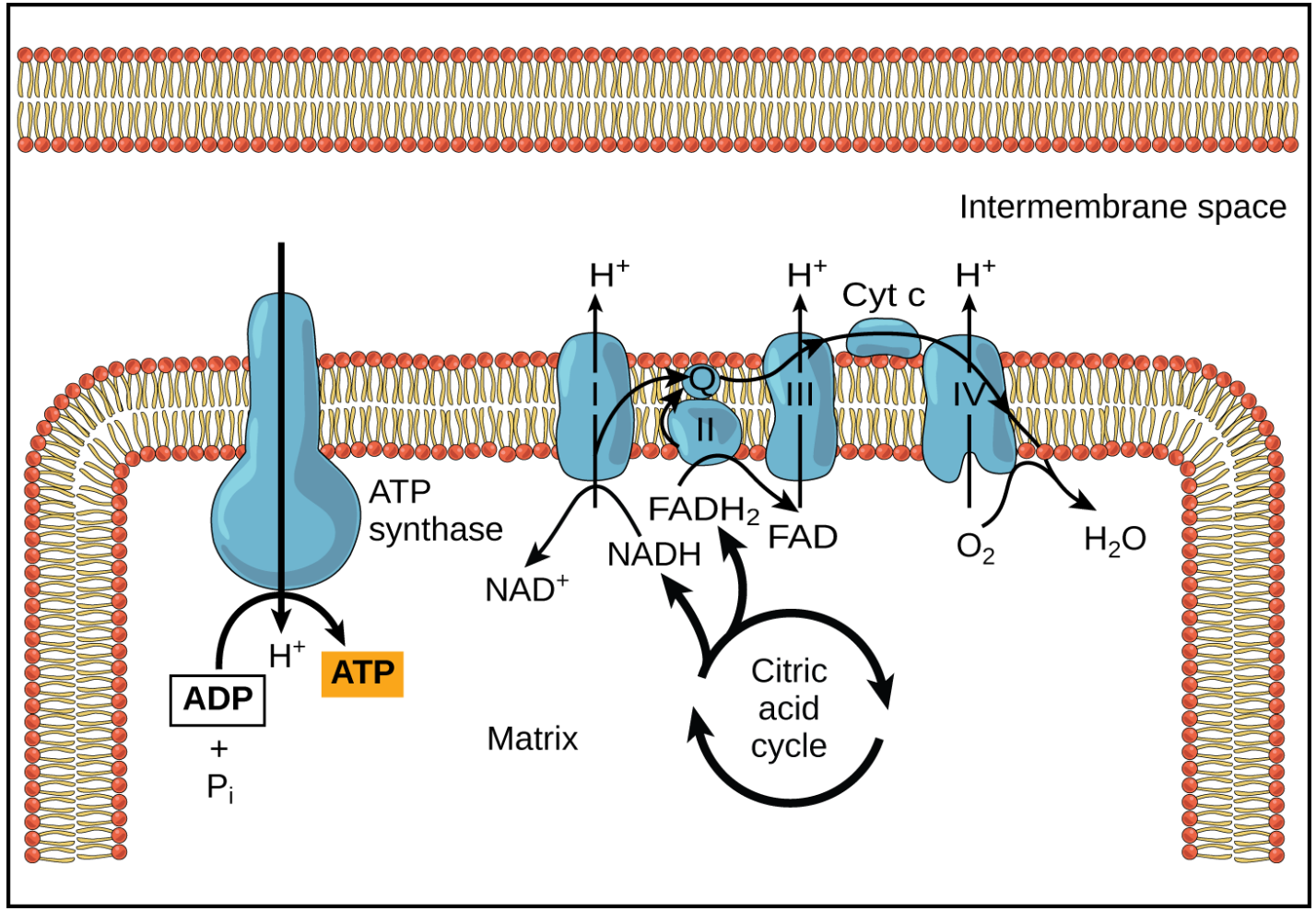
Product #1: _____ What happens to it? _____

Product #2: _____ What happens to it? _____

Product #3: _____ What happens to it? _____

Product #4: _____ What happens to it? _____

Cellular Respiration Overview: Oxidative Phosphorylation



1. H+ ions are transported from the mitochondrial _____ into the _____.

2. Is this active transport or passive transport?

3. If active transport, what provides the energy for this?

4. Beginning with NADH, track the flow of electrons through the ETC.

5. Beginning with $FADH_2$, track the flow of electrons through the ETC.

6. What is the final electron acceptor in the ETC?

7. What powers the production of ATP by ATP Synthase?

Cellular Poisons- Match the symptom to the poison

Poison: mechanism

- **Shikonin:** inhibits breakdown of glucose
- **Arsenic:** inhibits oxidation of pyruvate to acetyl co-A
- **Rotenone:** Disrupts Complex I to Ubiquinone
- **DNP:** makes mitochondrial inner membrane leaky to H+
- **Oligomycin:** Blocks ATP synthase

Symptom

A. Build up of H+ in intermembrane space. Very low ATP production.

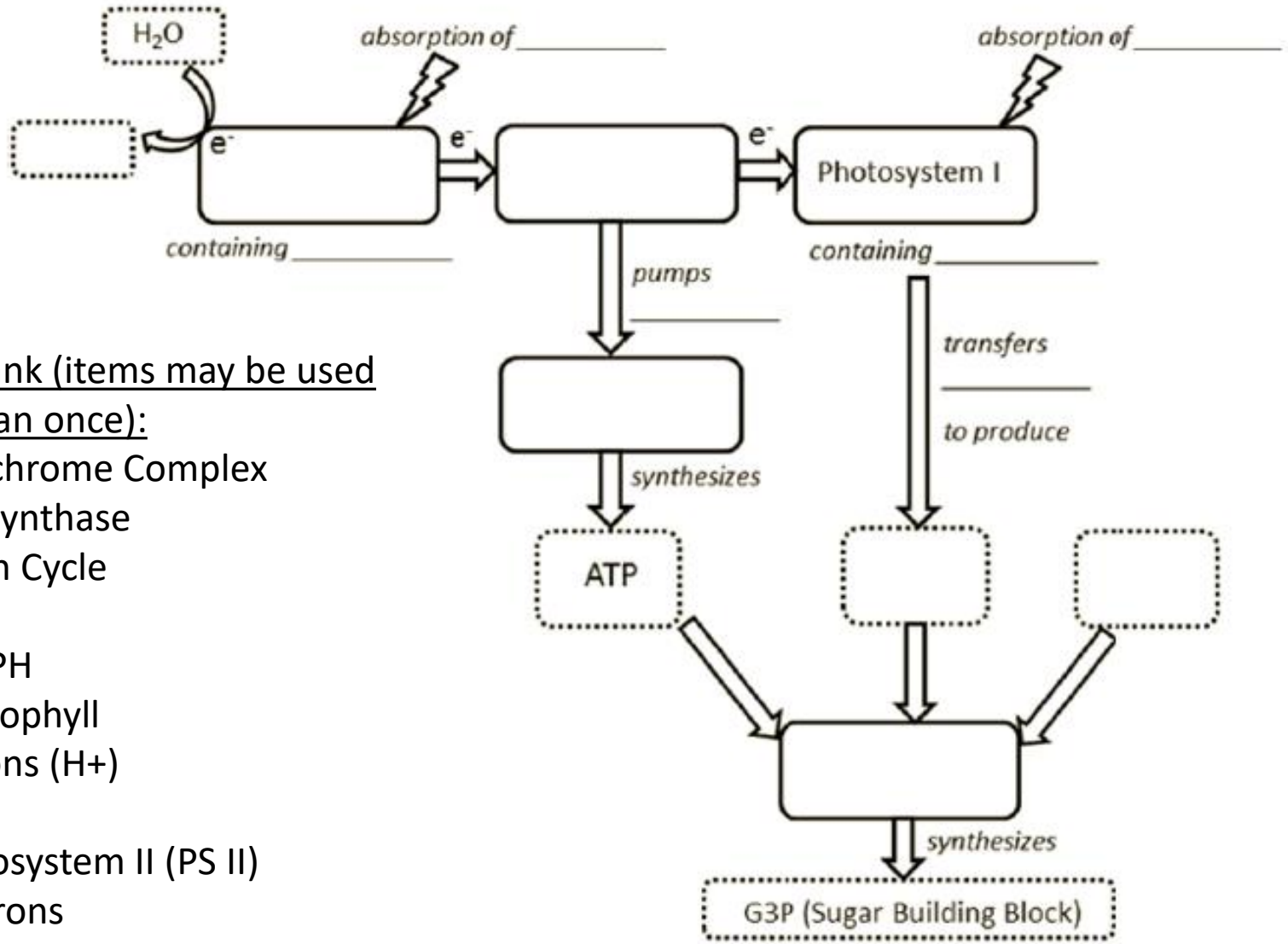
B. Extremely high glucose levels, CAC disabled, Oxidative Phosphorylation disabled, extremely low ATP production.

C. NADH levels increase in cell, $FADH_2$ levels are normal. ATP production is decreased.

D. Normal glucose levels. High level of pyruvate. CAC disabled. Extremely low ATP production.

E. Higher than normal levels of H+ in matrix. Lower than normal H+ in intermembrane space. Reduced ATP production.

#1: Fill in this concept map depicting the major steps in photosynthesis in the chloroplast:



<u>Word Bank (items may be used more than once):</u>

- Cytochrome Complex
- ATP Synthase
- Calvin Cycle
- Light
- NADPH
- Chlorophyll
- Protons (H+)
- CO_2
- Photosystem II (PS II)
- Electrons
- O_2

Herbicide	Chlorophyll Absorption	Oxygen Production	NADPH Production	ATP Synthase Activity	G3P Production
H-1	Normal	Normal	Decreased	Normal	Decreased
H-2	Decreased	Decreased	Decreased	Decreased	Decreased
H-3	Normal	Decreased	Decreased	Decreased	Decreased
H-4	Normal	Normal	Normal	Normal	Decreased
H-5	Normal	Normal	Normal	Decreased	Decreased

Each herbicide inhibits a specific part/step of photosynthesis. For each herbicide, list what it inhibits. Choose from the given word bank.

H-1:
H-2:
H-3:
H-4:
H-5:

Word Bank:
Chlorophyll absorption
Photosystem I
Photosystem II
ATP Synthase
Calvin Cycle